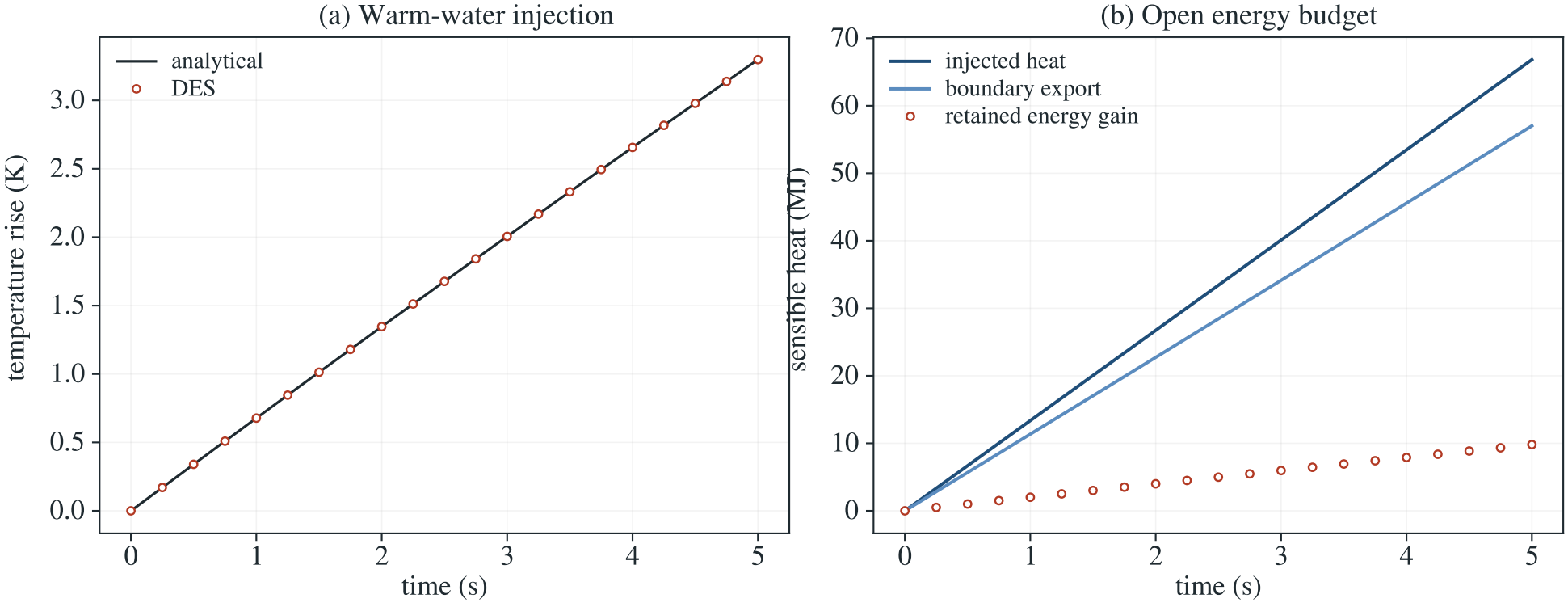

(a) Warm-water injection
analytical
DES
temperature rise (K)
time (s)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
1
2
3
4
5
(b) Open energy budget
injected heat
boundary export
retained energy gain
sensible heat (MJ)
time (s)
0
10
20
30
40
50
60
70